# ACI 318-19 RC-PN EXAMPLE 002

## Slab Punching Shear Design

### PROBLEM DESCRIPTION

The purpose of this example is to verify slab punching shear design in ETABS and SAFE.

The numerical example is a flat slab that has three 24-foot-long spans in each direction, as shown in Figure 1.

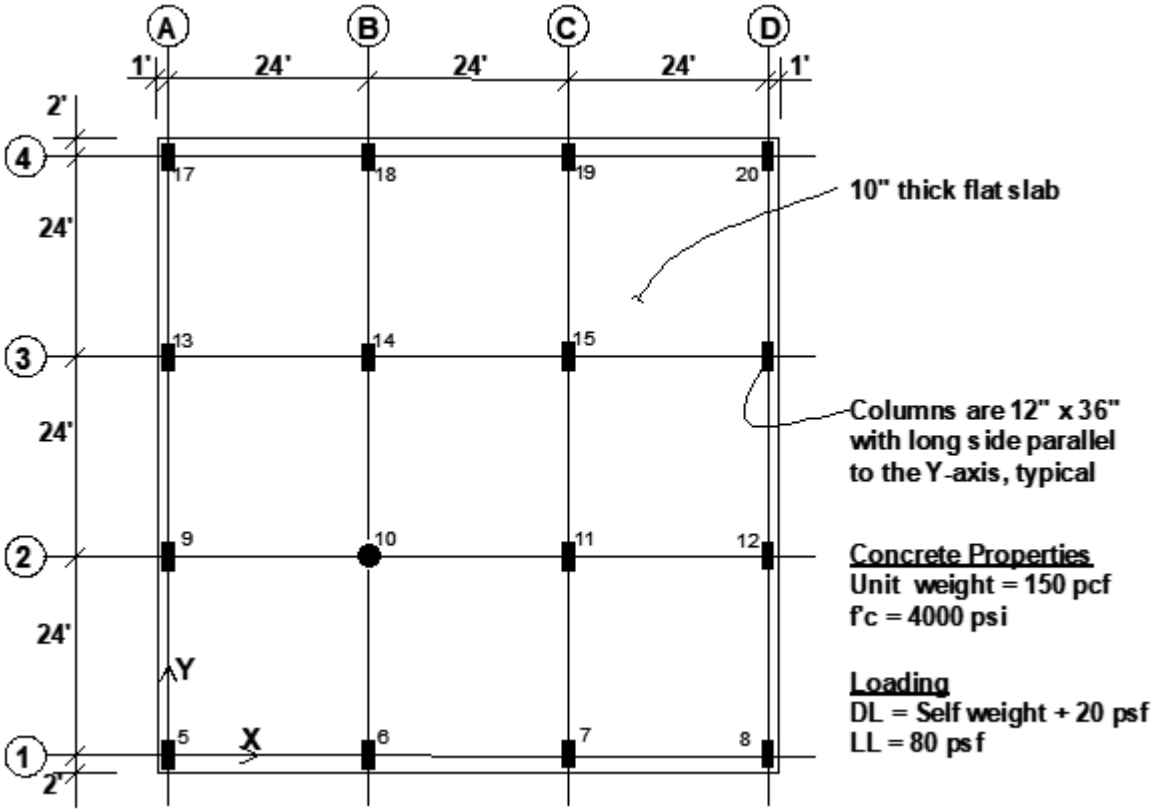



*Figure 1: Flat Slab For Numerical Example*

The slab overhangs the face of the column by 6 inches along each side of the structure. The columns are typically 12 inches wide by 36 inches long, with the long side parallel to the Y-axis except at grid line 2-B which has a circular column

of 24 inch diameter. The slab is typically 10 inches thick. Thick shell properties are used for the slab.
The concrete has a unit weight of 150 pcf and an f'c of 4000 psi. The dead load consists of the self weight of the structure plus an additional 20 psf. The live load is 80 psf.

## TECHNICAL FEATURES TESTED

- Calculation of punching shear capacity, shear stress, and D/C ratio.

## RESULTS COMPARISON

Table 1 shows the comparison of the punching shear capacity, shear stress ratio, and D/C ratio with the punching shear capacity, shear stress ratio and D/C ratio obtained by the analytical method. They match exactly for this example.

**Table 1 Comparison of Design Results for Punching Shear at Grid B-2**

| Method | Shear Stress (ksi) | Shear Capacity (ksi) | D/C ratio |
|---|---|---|---|
| ETABS, SAFE | 0.2730 | 0.190 | 1.44 |
| Calculated | 0.2730 | 0.190 | 1.44 |

## COMPUTER FILE: ACI 318-19 RC-PN Ex002

## CONCLUSION

The results show an exact comparison with the independent results.

## HAND CALCULATION

Hand Calculation for Interior Column

$d = [(10 - 1) + (10 - 2)] / 2 = 8.5"$

Refer to Figure 2.

$b_0 = 32.5\pi = 102.102"$

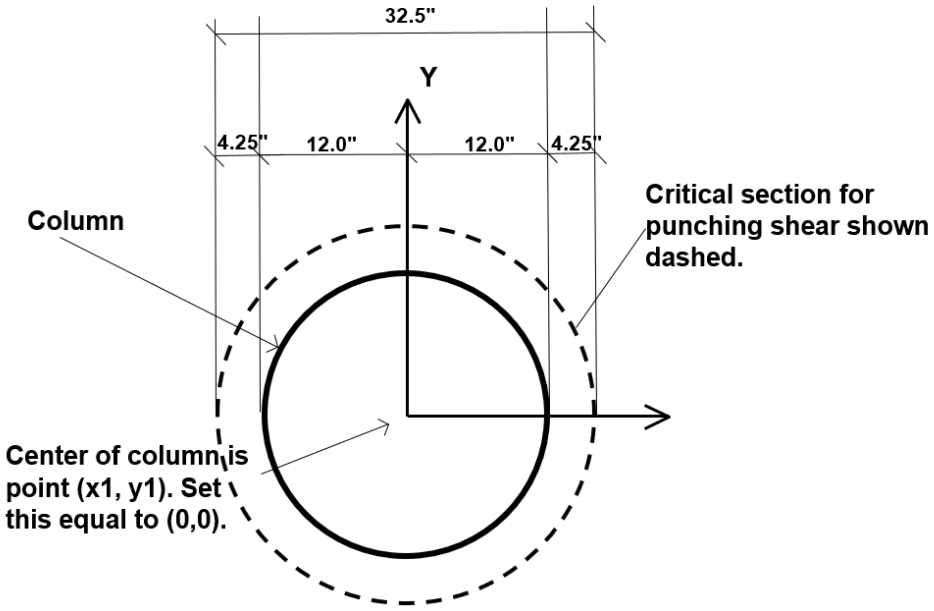



***Figure 2: Interior Column, Grid B-2 in Model***

$$\gamma_{V2} = 1 - \frac{1}{1 + \left(\frac{2}{3}\right)\sqrt{\frac{32.2}{32.2}}} = 0.4$$

$$\gamma_{V3} = 1 - \frac{1}{1 + \left(\frac{2}{3}\right)\sqrt{\frac{32.2}{32.2}}} = 0.4$$

The coordinates of the center of the column ($x_1$, $y_1$) are taken as (0, 0).

From the output at Grid B-2:

$V_U = 206.536$ k

$\gamma_{V2} M_{U2} = -84.049$ k-in

$\gamma_{V3} M_{U3} = 238.358$ k-in

To convert circular column dimension and punching perimeter into equivalent square:

Circular column area = $= \frac{\pi d^2}{4} = \frac{\pi (24)^2}{4} = 452.389 \text{ in}^2$

Equivalent square column area = $= c^2$

$c = \sqrt{452.389} = 21.3"$

Similarly, punching perimeter, $b_0 = 32.5\pi = 102.102"$ is converted into equivalent square:

$b_o = 4c_X = 4c_Y = 102.102"$

$c_X = c_Y = 25.526"$

The basic equation for the ACI 318-19 is:

$$v_U = \frac{V_U}{A_C} + \frac{\gamma_{VX} M_{UX} C_X}{J_X} + \frac{\gamma_{VY} M_{UY} C_Y}{J_Y} \text{ s}$$

$A_C = b_o \, d = 102.102 * 8.5 = 867.865 \text{ in}^2$

$$J_X = J_Y = \frac{d(c_X + d)^3}{6} + \frac{(c_X + d)d^3}{6} + \frac{d(c_X + d)^3}{2}$$

$$J_X = J_Y = \frac{8.5(25.526)^3}{6} + \frac{(25.526)8.5^3}{6} + \frac{8.5(25.526)^3}{2} = 96{,}861.5211 \, in^4$$

$$c_X = c_Y = \frac{c}{2} = \frac{21.3}{2} = 10.65"$$

$$v_U = \frac{206.536}{867.865} + \frac{0.4 \bullet 210.1236 \bullet 10.65}{96{,}861.5211} + \frac{0.4 \bullet 595.8984 \bullet 10.65}{96{,}861.5211}$$

$$v_U = 0.238 + 0.009 + 0.026 = 0.273 \text{ ksi}$$

The shear capacity is calculated based on the smallest of ACI 318-19 equations (a), (b), and (c) of Table 22.6.5.2 with the $b_0$ and d terms removed to convert force to stress. In the equations,

$$\lambda = 1.0$$

$$\lambda_s = \sqrt{\frac{2}{1+\frac{d}{10}}} = \sqrt{\frac{2}{1+\frac{8.5}{10}}} = 1.0397 \le 1.0, \quad \lambda_s = 1.0$$

$$\varphi v_C = \frac{0.75 \bullet 4 \bullet \lambda_s \lambda \sqrt{4000}}{1000} = 0.190 \text{ ksi in accordance with equation (a)}$$

$$\varphi v_C = \frac{0.75\left(2+\frac{4}{24/24}\right)\lambda_s \lambda \sqrt{4000}}{1000} = 0.296 \text{ ksi in accordance with equation (b)}$$

$$\varphi v_C = \frac{0.75\left(2+\frac{40 \bullet 8.5}{130}\right)\lambda_s \lambda \sqrt{4000}}{1000} = 0.219 \text{ ksi in accordance with equation (c)}$$

Equation (b) of ACI 318-19 Table 22.6.5.2 yields the smallest value of $\boxed{\phi v_C = 0.190 \text{ ksi}}$ and thus this is the shear capacity.

$$\boxed{\textit{Shear Ratio} = \frac{v_U}{\varphi v_C} = \frac{0.273}{0.190} = 1.44}$$